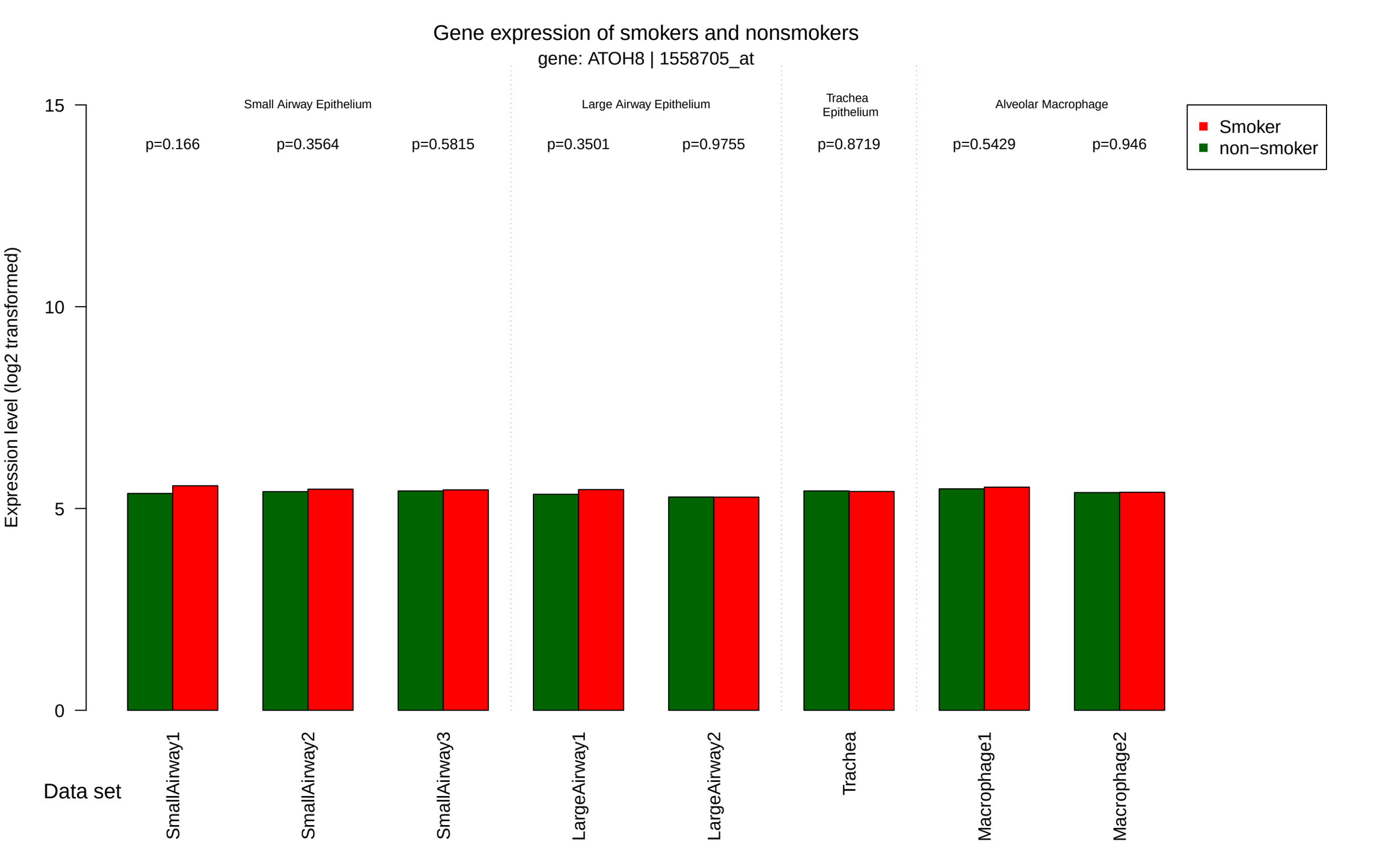
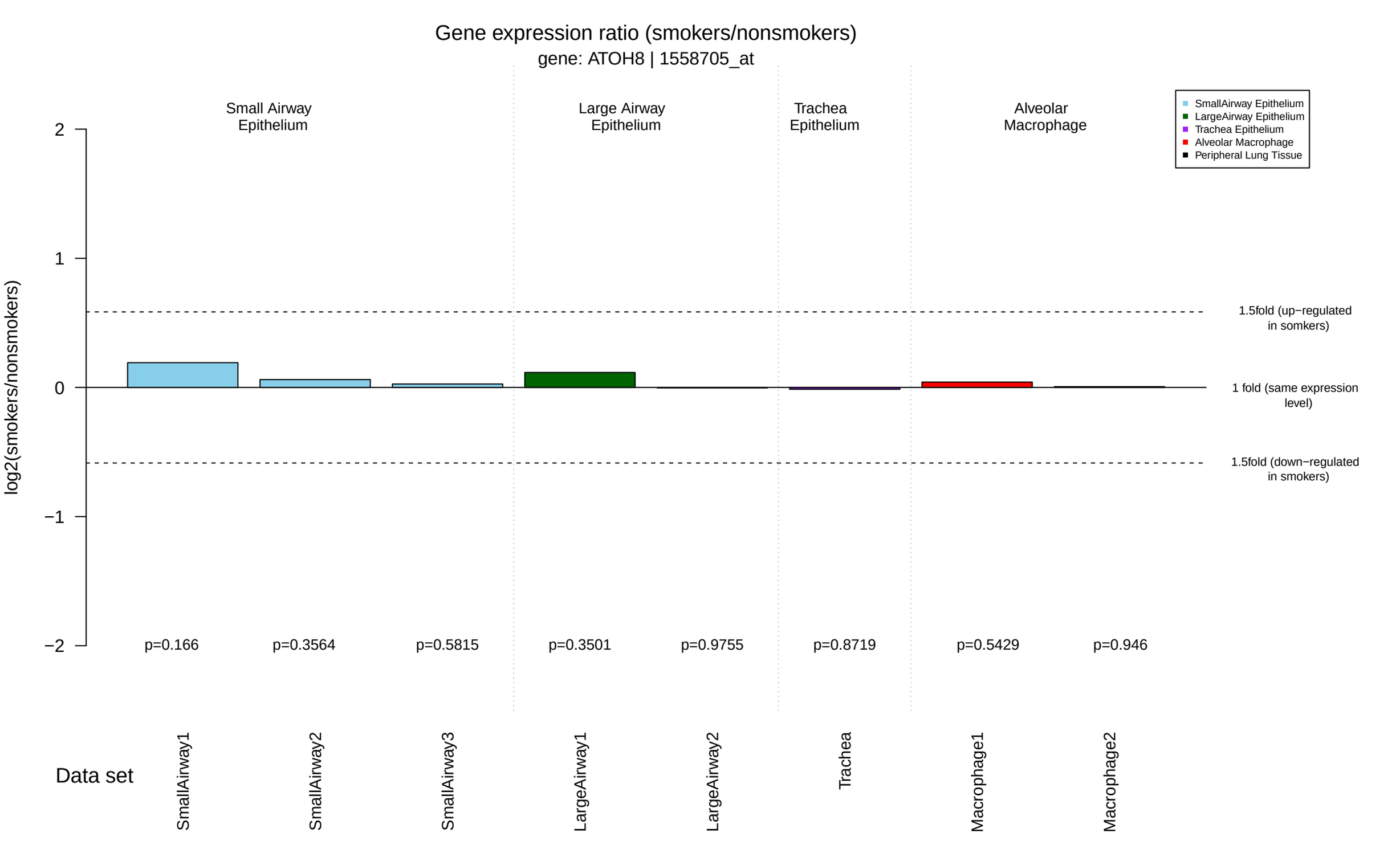
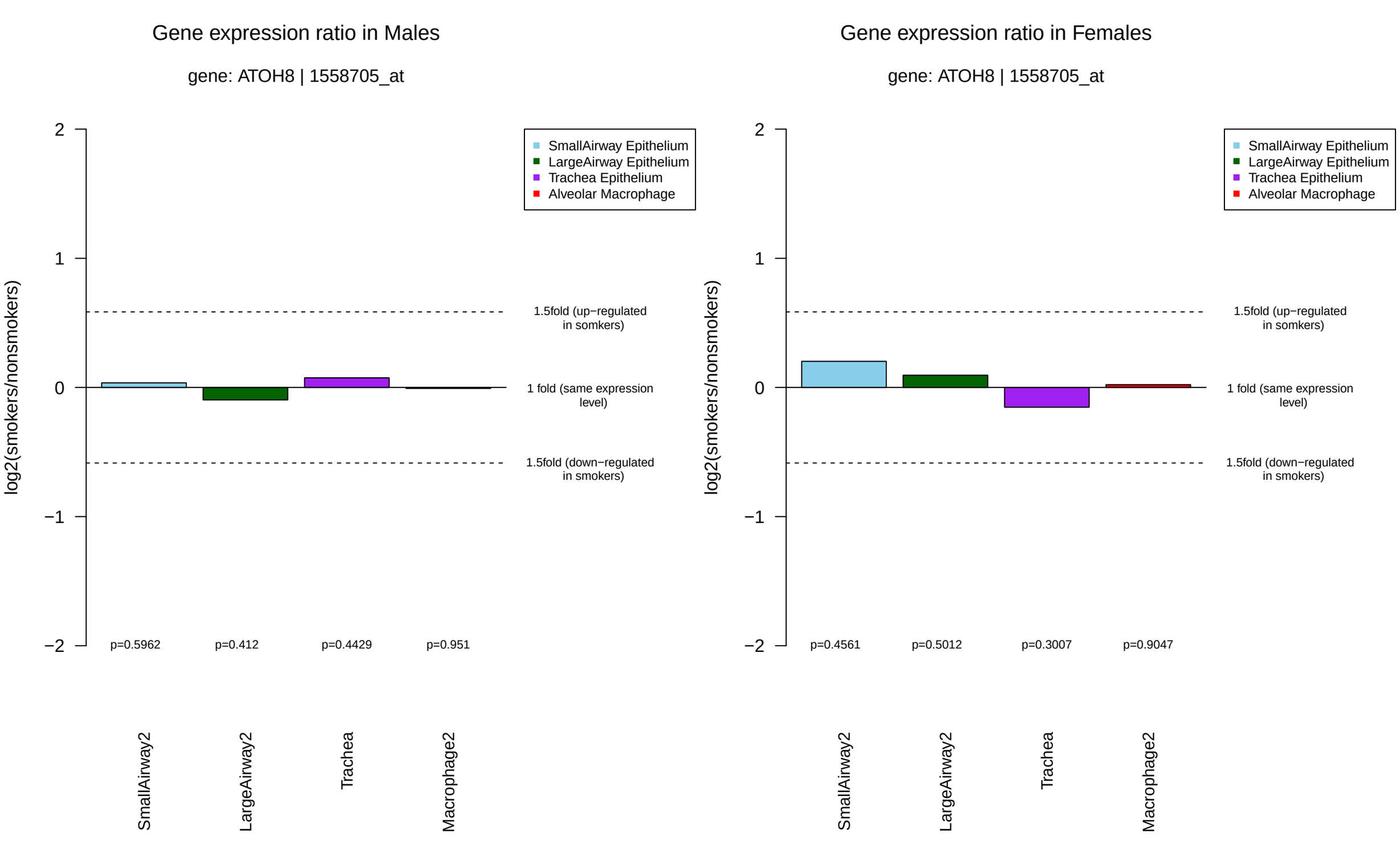
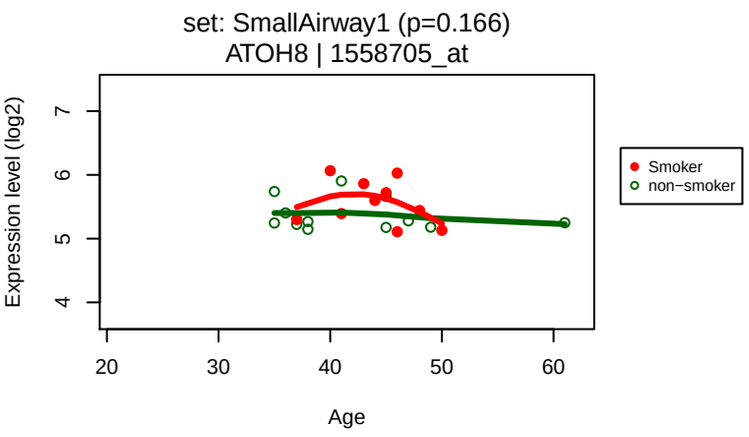
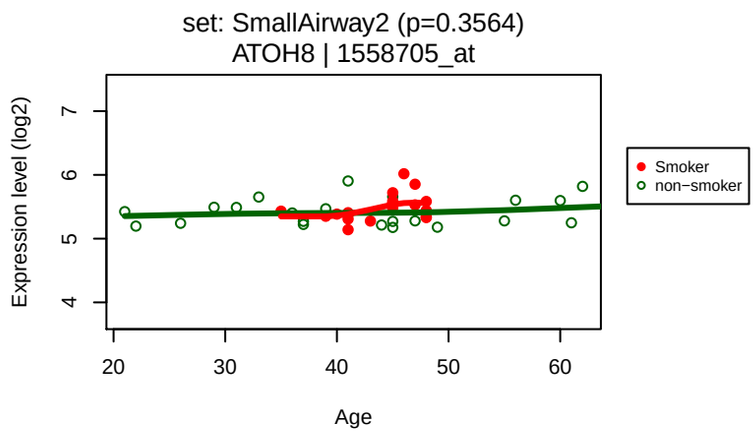
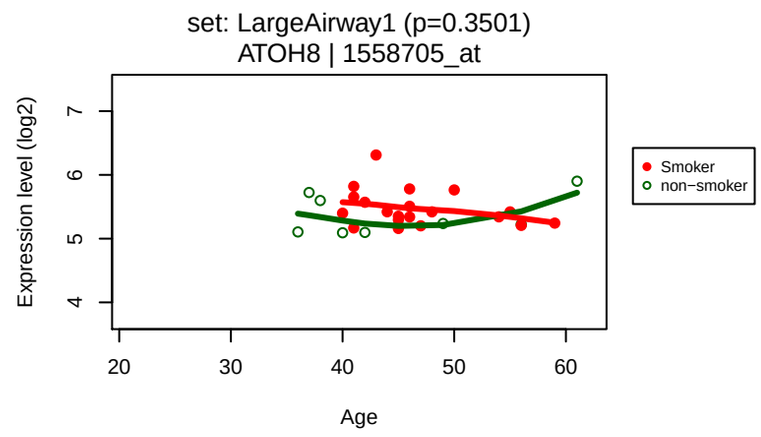

Gene expression of smokers and nonsmokers
gene: ATOH8 | 1558705_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.166
p=0.3564
p=0.5815
p=0.3501
p=0.9755
p=0.8719
p=0.5429
p=0.946
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

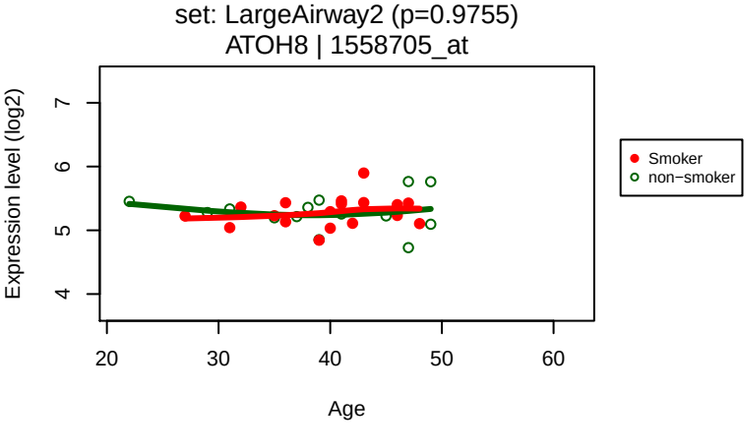

Gene expression ratio (smokers/nonsmokers)
gene: ATOH8 | 1558705_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.166
p=0.3564
p=0.5815
p=0.3501
p=0.9755
p=0.8719
p=0.5429
p=0.946
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

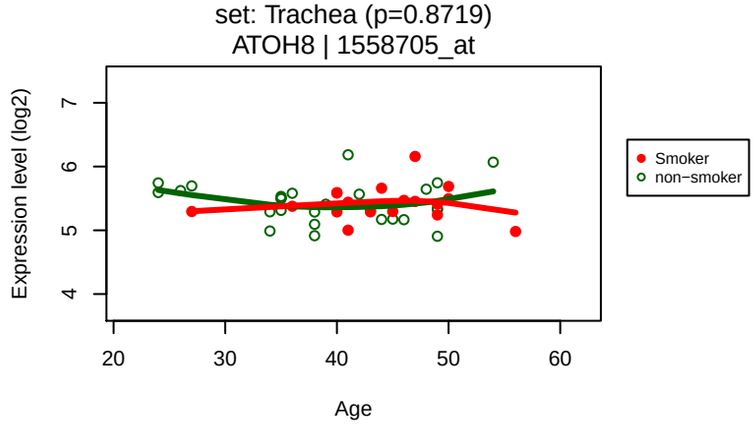

Gene expression ratio in Males
gene: ATOH8 | 1558705_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5962
p=0.412
p=0.4429
p=0.951
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATOH8 | 1558705_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4561
p=0.5012
p=0.3007
p=0.9047
SmallAirway2
LargeAirway2
Trachea
Macrophage2

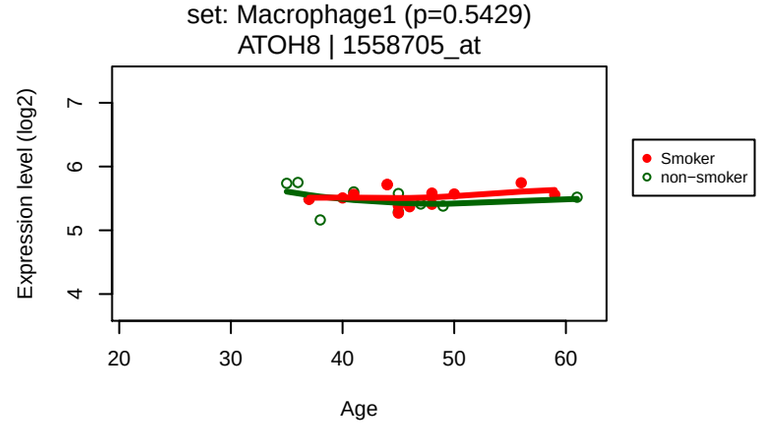

set: SmallAirway1 (p=0.166)
ATOH8 | 1558705_at
Expression level (log2)
Age
Smoker
non−smoker

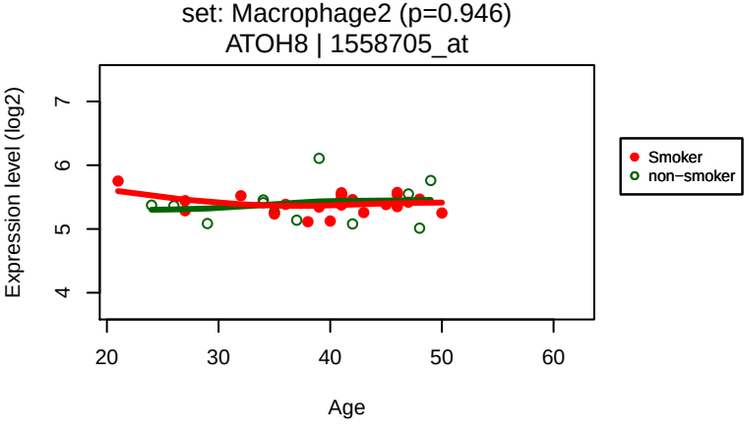

set: SmallAirway2 (p=0.3564)
ATOH8 | 1558705_at
Expression level (log2)
Age
Smoker
non−smoker

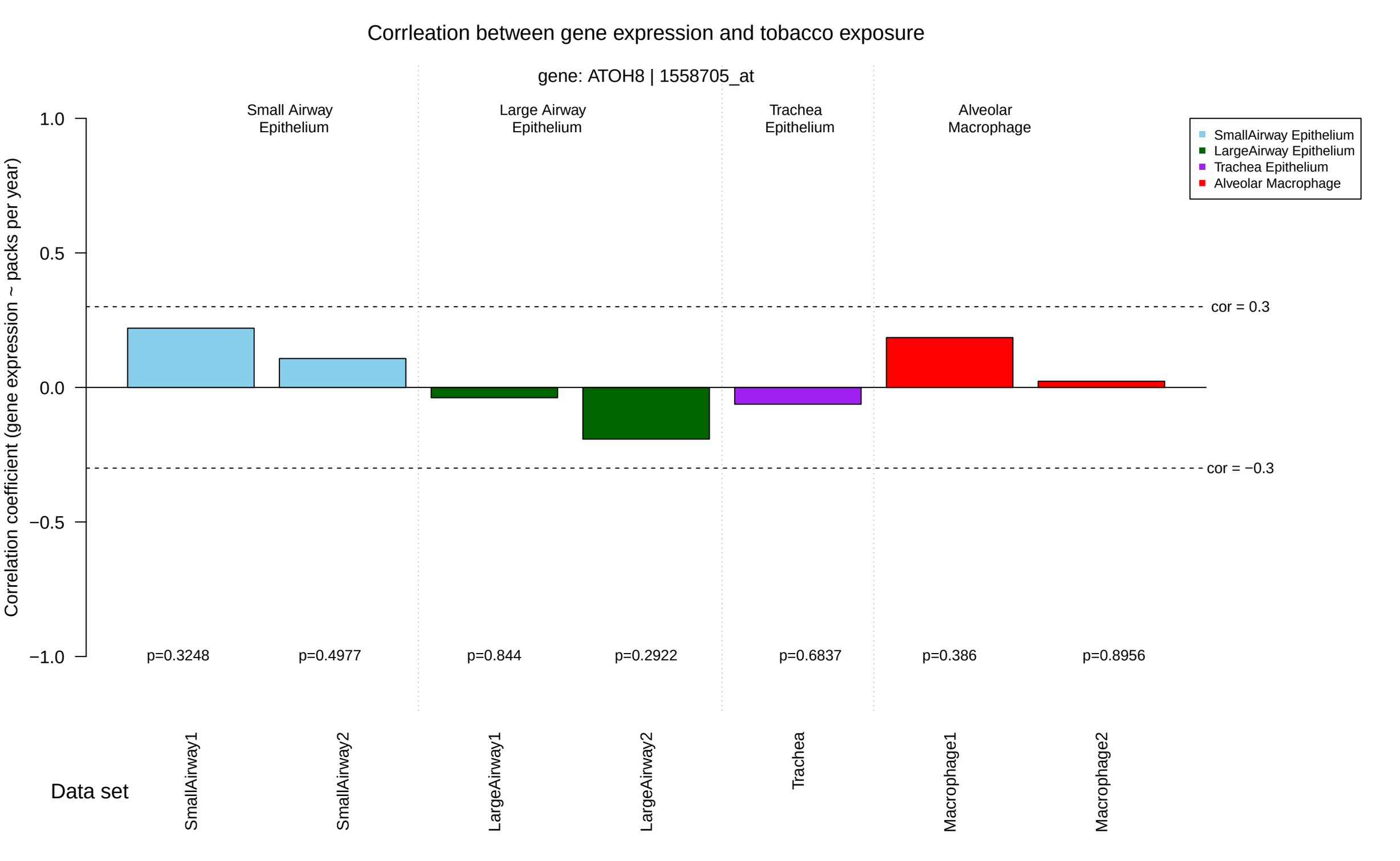

set: LargeAirway1 (p=0.3501)
ATOH8 | 1558705_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9755)
ATOH8 | 1558705_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8719)
ATOH8 | 1558705_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5429)
ATOH8 | 1558705_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.946)
ATOH8 | 1558705_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATOH8 | 1558705_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3248
p=0.4977
p=0.844
p=0.2922
p=0.6837
p=0.386
p=0.8956
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2